Li-Fi (Light Fidelity): A Critical Review of Operating Principles, Modulation, Standards, Performance Claims and Open Challenges

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Abstract

Li-Fi (Light Fidelity) uses light from light-emitting diodes (LEDs), laser diodes or infrared sources as the medium for high-speed, bidirectional and networked wireless communication. Because the optical spectrum is unlicensed, free of radio-frequency (RF) interference and confined by opaque walls, Li-Fi is studied as a complement to Wi-Fi and cellular networks. This review (i) separates the terminology of visible light communication (VLC), Li-Fi and free-space optics (FSO); (ii) describes the system architecture and the intensity-modulation/direct-detection channel; (iii) surveys single-carrier, multi-carrier and color-domain modulation; (iv) summarizes the IEEE 802.15.7, ITU-T G.9991 and IEEE 802.11bb standards; (v) places widely quoted data rates in context by separating laboratory demonstrations from standardized and commercial performance; (vi) assesses applications and their maturity; and (vii) discusses limitations and research directions. Li-Fi is best understood as a short-range, high-capacity complement to RF rather than a replacement. Standardization has moved it from laboratory demonstration toward early commercial deployment, while uplink design, multi-cell mobility and client device support remain open issues.

Keywords: Li-Fi, Visible light communication, Optical wireless communication, IEEE 802.11bb, Optical OFDM, Hybrid networks

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1. Introduction

Wireless data traffic continues to grow while the radio-frequency (RF) spectrum that carries most of it is finite and increasingly shared. The optical spectrum, by contrast, offers a very large unlicensed bandwidth that conventional wireless systems do not use (Haas 2018; Haas et al. 2016; Haas et al. 2020). Li-Fi exploits this resource: the same light-emitting diodes (LEDs) that illuminate a room are modulated faster than the eye can perceive, and a photodetector in the user's device recovers the data. In the terminology that has become standard, Li-Fi denotes a complete, bidirectional and networked wireless system built on this principle, and not merely a point-to-point optical link (Haas et al. 2016).

Four properties are usually offered as motivation. The first is *capacity*: optical cells are small, so spectrum can be reused densely, which suggests high aggregate capacity per unit area (Haas 2018; Haas et al. 2016). The second is *accessibility*: light does not cause RF interference, which is attractive in settings where radio emissions are restricted or undesirable, such as aircraft cabins and some clinical areas (Air France 2019; Johnson 2023). The third is *security*: because light does not pass through walls, the signal is physically confined to the illuminated space. The fourth is *infrastructure reuse*: existing lighting points can serve as access points. Each of these claims carries qualifications, which this review examines rather than assumes. For example, confinement reduces but does not eliminate eavesdropping risk, and the energy benefit of reusing luminaires has to be weighed against the power drawn by the additional front-end electronics.

This article is a narrative review. It is an expanded and updated version of a 2025 research article prepared as the course project for Research and Presentation Methods, revised in October 2026 to incorporate standardization and performance results that the original report did not cover. The contributions are fourfold. First, the terminology of visible light communication (VLC), Li-Fi and free-space optics (FSO) is separated, since these terms are often conflated. Second, the standards landscape is summarized, including IEEE 802.11bb (2023), which defines a Li-Fi amendment to the Wi-Fi standard. Third, headline data rates are reported with their experimental conditions so that laboratory records are not mistaken for product performance. Fourth, applications and limitations are assessed together with an indication of maturity. Sources were selected for relevance, with preference given to peer-reviewed journals, standards documents and primary institutional publications. Vendor and press material is used only where no primary source was found, and it is identified as such in the text. Sources were identified through keyword searches (for example Li-Fi, visible light communication, IEEE 802.11bb, optical OFDM and free-space optics) and by following citations from key papers. No formal systematic-review protocol was applied, so coverage should be regarded as selective rather than exhaustive. Web sources were last accessed in October 2026. In line with Springer Nature policy, the use of generative AI is documented here: the AI assistant Claude (Anthropic) was used to help check statements against sources, restructure the text, edit the language and generate the figure files. The author reviewed all content and takes full responsibility for it.

The remainder of the article is organized as follows. Section 2 gives background and terminology. Section 3 describes the system architecture and channel. Section 4 surveys modulation and multiple-access techniques. Section 5 summarizes standards, and Section 6 interprets reported data rates. Section 7 reviews applications, Section 8 discusses limitations, Section 9 covers the related technology of free-space optics, Section 10 outlines research directions, and Section 11 concludes.

2. Background and Terminology

2.1 From visible light communication to Li-Fi

Research on communication using white LED lighting developed in the early 2000s, notably at Keio University in Japan, where indoor transmission using multiple white LED luminaires and the basic channel characteristics were analyzed (Komine and Nakagawa 2004; Tanaka et al. 2001). A first IEEE standard for short-range optical wireless communication, IEEE 802.15.7, followed in 2011 and was revised in 2018 (IEEE 2018). The term *Li-Fi* is generally attributed to Harald Haas, who demonstrated data transmission from an LED lamp at a TED Global talk in 2011 and later co-founded pureLiFi. In the sense used by Haas and co-authors, VLC refers to point-to-point communication using visible light, whereas Li-Fi refers to the networked case, with multiple access points, multiple users and handover between cells (Haas et al. 2016). This distinction matters for the rest of this review: a laboratory VLC link can demonstrate a record data rate without being a Li-Fi system. Book-length treatments of the underlying principles are available in (Dimitrov and Haas 2015).

2.2 Optical bands

Visible light occupies roughly 380–750 nm. Many practical Li-Fi products use near-infrared (NIR) light for part or all of the link, because it is invisible, does not depend on the room being lit, and avoids illumination constraints such as dimming and flicker. The first commercial module compliant with IEEE 802.11bb, for instance, is reported to operate at 850 nm (CNX Software 2023). Earlier accounts of Li-Fi that describe it as dependent on visible illumination therefore apply to one family of implementations and not to the technology as a whole.

2.3 Li-Fi in the optical wireless landscape

Li-Fi belongs to the broader family of optical wireless communication (OWC). Table 1 contrasts it with Wi-Fi and with FSO links, which use narrow laser beams between fixed terminals over long distances. The entries are indicative and are intended to show the different roles of the three technologies rather than to rank them.

Table 1 Indicative comparison of Wi-Fi, Li-Fi and free-space optical (FSO) links

Aspect	Wi-Fi (RF)	Li-Fi	FSO point-to-point
Medium	Radio waves (2.4, 5, 6 GHz bands)	Visible and/or near-infrared light from LEDs or laser diodes	Narrow, typically invisible laser beams
Typical range	Tens of meters indoors	Meters per cell; the first 802.11bb module is specified for about 20–300 cm (CNX Software 2023)	Up to about 20 km (commercial Lightbridge) (Krishnaswamy 2025)
Passes through walls	Yes	No	No
Mobility	Mature roaming	Requires multi-cell handover; still maturing (Haas et al. 2016)	Fixed terminals, alignment required
Main impairments	Spectrum sharing, RF interference	Blockage, ambient light, limited LED bandwidth	Fog, rain, turbulence, misalignment
Typical role	General-purpose coverage	Complement in dense, RF-sensitive or secure indoor areas	Backhaul where fiber is impractical

3. System Architecture and Channel

3.1 Architecture

Fig. 1 shows a generic Li-Fi link. Data from a server or the Internet reach a Li-Fi access point through a wired backhaul, preferably power-over-Ethernet (PoE) in new installations or power-line communication (PLC) in retrofits (Claburn 2023). Inside the access point, a modulator and driver vary the optical output of an LED or laser diode. At the user terminal a photodetector converts the received optical power into a photocurrent, a transimpedance amplifier (TIA) and demodulator recover the data stream, and the result is passed to the device. A networked system also requires an uplink, which is commonly provided by a separate infrared transmitter or, in hybrid designs, by an RF link (Haas et al. 2016).

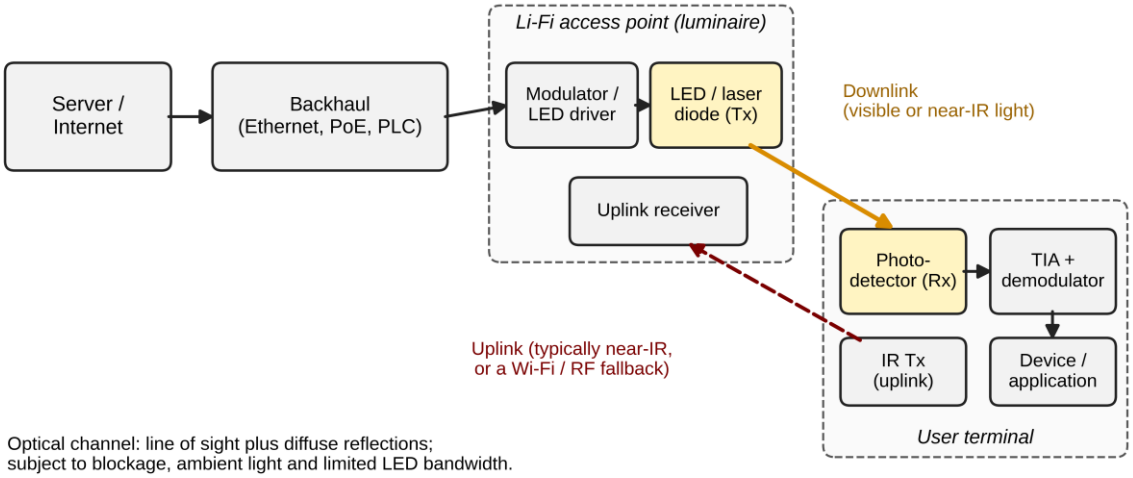


Fig. 1 Generic Li-Fi architecture with a downlink in visible or near-infrared light and an uplink that is typically infrared or RF

3.2 The optical channel

Li-Fi uses intensity modulation with direct detection (IM/DD). The transmitted signal is the instantaneous optical power $x(t)$, which must be real and non-negative, and the receiver responds to power and not to field amplitude. This is the origin of several design choices in Section 4, such as the DC bias in optical orthogonal frequency division multiplexing (OFDM) (Armstrong and Lowery 2006; Islim and Haas 2016). The received photocurrent for a line-of-sight (LOS) link can be written as

$$y(t) = R \cdot H(0) \cdot x(t) + n(t), \quad 0 \le x(t) \le P_{\text{peak}} \tag{1}$$

where R is the photodetector responsivity, $n(t)$ is noise (shot noise from signal and ambient light, plus thermal noise), and P_{peak} is the peak optical power. For a Lambertian emitter, the LOS DC channel gain is commonly modeled as (Komine and Nakagawa 2004)

$$H(0) = [(m+1)A / (2\pi d^2)] \cdot \cos^{m}(\varphi) \cdot T_s(\psi) \cdot g(\psi) \cdot \cos\psi, \quad 0 \le \psi \le \Psi_c \tag{2}$$

$$m = -\ln 2 / \ln(\cos \Phi_{1/2}) \tag{3}$$

Here A is the detector area, d the distance, φ the irradiance angle, ψ the incidence angle, $T_s(\psi)$ the optical filter gain, $g(\psi)$ the concentrator gain, Ψ_c the receiver field of view, and $\Phi_{1/2}$ the half-power semi-angle of the LED. Equation (2) captures the two facts that matter most in practice: received power falls with the square of distance, and it depends strongly on the alignment between the luminaire and the receiver. Reflections from walls and furniture add a non-line-of-sight (NLOS) component, but this is typically

much weaker than the LOS term, so blockage of the direct path can degrade a link sharply (Komine et al. 2005).

3.3 Practical constraints

The modulation bandwidth of commercial phosphor-converted white LEDs is limited, which is the main bottleneck for data rate. Gallium-nitride micro-LEDs have been demonstrated with modulation bandwidths of at least 60 MHz, substantially higher than commercial lighting LEDs (Tsonev et al. 2014), and have supported multi-gigabit transmission (Ferreira et al. 2016; Tsonev et al. 2014). Because the same LED provides illumination, the communication scheme must also support dimming and avoid visible flicker, which the IEEE 802.15.7 standard addresses through its dimming mechanisms (IEEE 2018). Vendors report that their equipment continues to operate when luminaires are dimmed to about 10% of maximum output (Claburn 2023), and NIR operation removes the dependence on illumination altogether.

4. Modulation, Multiplexing and Multiple Access

Modulation schemes originally developed for RF can be adapted to Li-Fi, subject to the real and non-negative constraint of IM/DD (Devi et al. 2023; Islim and Haas 2016). Fig. 2 organizes the main families, and Table 2 summarizes their trade-offs.

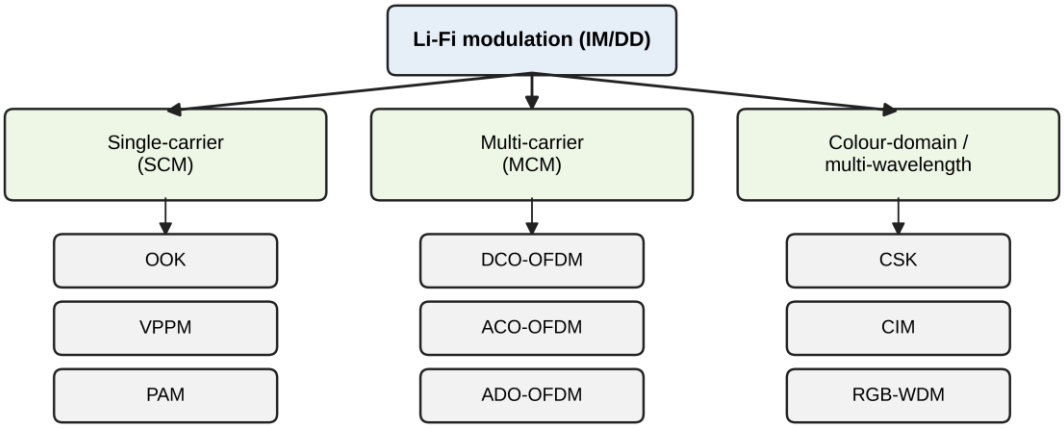


Fig. 2 Taxonomy of Li-Fi modulation schemes

4.1 Single-carrier modulation

On-off keying (OOK) represents bits by switching the LED between two intensity levels. It is the simplest scheme and is the baseline in IEEE 802.15.7. The standard uses Manchester coding for low rates, which doubles the bandwidth required, and run-length-limited (RLL) codes for higher rates, which are more spectrally efficient; dimming is supported by adjusting the duty cycle or intensity of the OOK waveform (IEEE 2018; Islim and Haas 2016).

Variable pulse position modulation (VPPM) combines pulse width modulation (PWM) with binary pulse position modulation (PPM). The pulse width sets the dimming level, while the pulse position carries the data and keeps the average brightness constant, which prevents visible flicker (Lee et al. 2016; Yoo and Jung 2013).

Pulse amplitude modulation (PAM) uses several intensity levels per symbol to increase spectral efficiency. Micro-LEDs have been shown to support PAM and OFDM transmission at rates up to 5 Gb/s (Ferreira et al. 2016).

4.2 Multi-carrier modulation

OFDM divides the signal over many parallel subcarriers. It provides high spectral efficiency, resists inter-symbol interference caused by the limited LED bandwidth, and allows adaptive bit loading, which was used to exceed 3 Gb/s with a single micro-LED (Tsonev et al. 2014). Because IM/DD requires a real, non-negative signal, optical variants are used. In DC-biased optical OFDM (DCO-OFDM) a DC bias is added so that the signal is almost always positive, at the cost of power efficiency. In asymmetrically clipped optical OFDM (ACO-OFDM) only odd subcarriers carry data and the negative part is clipped without information loss, which removes the bias but halves the spectral efficiency (Armstrong and Lowery 2006). Hybrid variants such as asymmetrically clipped DC-biased optical OFDM (ADO-OFDM) combine the two approaches (Dissanayake and Armstrong 2013).

4.3 Color-domain and multi-wavelength techniques

With red-green-blue (RGB) LEDs, information can be placed in the color of the light. In **color shift keying (CSK)** the chromaticity changes while the total intensity stays approximately constant, which avoids flicker but requires more complex transmitter and receiver designs (Islim and Haas 2016). **Color intensity modulation (CIM)** varies the intensities of the individual colors while the perceived color remains constant, and is generally regarded as a simplified relative of CSK (Devi et al. 2023; Islim and Haas 2016). Independent data streams can also be sent on different wavelengths using wavelength division multiplexing (WDM), the approach behind the highest laboratory data rates discussed in Section 6 (Gomez et al. 2015).

4.4 Multiple access

Within a cell, several users can share the optical resource using time, frequency or code division, and non-orthogonal multiple access (NOMA) has been analyzed for VLC downlinks (Yin et al. 2016). Across cells, spatial reuse is the main source of capacity, which makes cell planning and handover central to a Li-Fi network (Haas et al. 2016; Matheus et al. 2019).

Table 2 Summary of Li-Fi modulation schemes

Scheme	Principle	Strength	Limitation
OOK	Two intensity levels	Simple; baseline of IEEE 802.15.7 (IEEE 2018)	Manchester coding doubles the bandwidth; low spectral efficiency
VPPM	PWM combined with binary PPM	Dimming control without visible flicker (Yoo and Jung 2013)	Modest data rates
PAM	Several intensity levels	Higher spectral efficiency than OOK (Ferreira et al. 2016)	Sensitive to noise and LED non-linearity
DCO-OFDM	OFDM with DC bias	High spectral efficiency; adaptive bit loading (Tsonev et al. 2014)	Bias reduces power efficiency
ACO-OFDM	Odd subcarriers only, clipped	No DC bias; power efficient (Armstrong and Lowery 2006)	Half the spectral efficiency of DCO-OFDM
ADO-OFDM	ACO and DCO combined	Intermediate trade-off (Dissanayake and Armstrong 2013)	Higher complexity
CSK / CIM	Data in color or color intensities	Constant perceived intensity or color (Islim and Haas 2016)	Complex RGB transmitter and receiver design
WDM	Parallel streams on separate wavelengths	Scales the rate with the number of channels (Gomez et al. 2015)	Multiple sources, filters and alignment hardware

5. Standards

Three standards frame the field (Table 3). IEEE 802.15.7, first published in 2011 and revised in 2018, covers short-range optical wireless communication, including physical-layer (PHY) modes and dimming support (IEEE 2018). ITU-T G.9991, approved in 2019, specifies the system architecture, physical layer and data-link layer of a high-speed indoor VLC transceiver, with a physical layer based on the G.9960 family. Its Amendment 1 (2020) adds IEEE 802.1X-based authentication and Amendment 2 (2021) adds a mechanism for inter-domain mobility through an external controller (ITU-T 2019).

The most consequential recent development is IEEE 802.11bb, the light communications amendment to the Wi-Fi standard. Its task group was formed in 2018, chaired by pureLiFi and supported by Fraunhofer HHI, and the standard was ratified in June 2023 (IEEE ComSoc Technology Blog 2023). It defines a light-communications PHY together with the medium access control (MAC) changes needed to operate it, with the stated aim that light can appear to the system as another Wi-Fi band and that Li-Fi can interoperate with Wi-Fi equipment from multiple vendors (IEEE ComSoc Technology Blog 2023; pureLiFi 2023). Its theoretical peak throughput is 9.6 Gb/s, the same nominal figure as Wi-Fi 6 (Claburn 2023; pureLiFi 2023). Some descriptions state that 802.11bb covers both visible and infrared light (Claburn 2023), whereas the task-group vendors' materials and the first compliant hardware emphasize near-infrared operation (CNX Software 2023; pureLiFi 2023). The practical consequence is that "Li-Fi" in the 802.11 sense should not be equated with illumination-based communication.

Table 3 Standards relevant to Li-Fi

Standard	Year	Scope	Notes
IEEE 802.15.7	2011; revised 2018	Short-range optical wireless communication	PHY modes with OOK, VPPM and others; dimming support (IEEE 2018)
ITU-T G.9991	2019; Amd. 1 (2020), Amd. 2 (2021)	High-speed indoor VLC transceiver: architecture, PHY and data-link layer	PHY based on G.9960; 802.1X authentication; inter-domain mobility (ITU-T 2019)
IEEE 802.11bb	2023	Light communications amendment to Wi-Fi (PHY and MAC)	Nominal peak 9.6 Gb/s; near-infrared focus; interoperability with the Wi-Fi ecosystem (Claburn 2023; IEEE 2023; pureLiFi 2023)

6. Data Rates in Context

Li-Fi literature and promotional material quote data rates that span more than two orders of magnitude. Fig. 3 places the most frequently cited figures side by side. They are not directly comparable, because they differ in light source, wavelength, distance, receiver mobility and whether the system is a laboratory link, a standard or a product.

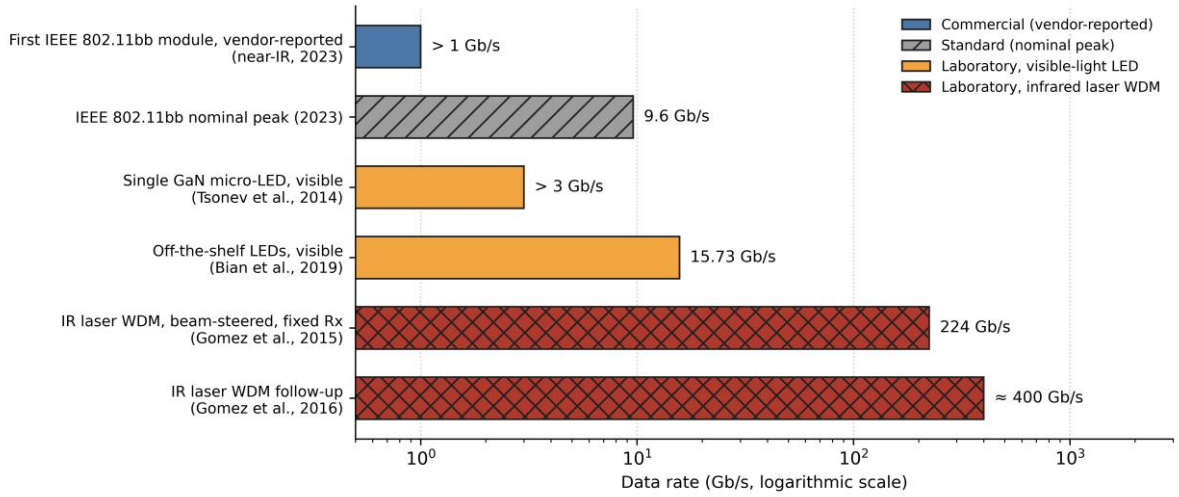


Fig. 3 Reported data rates for Li-Fi and related optical wireless links. Values are taken from the cited sources; ">" indicates a reported lower bound. Laboratory links, nominal standard peaks and vendor-reported products are not directly comparable

Visible-light LED links. A single 50 µm gallium-nitride micro-LED has been used to demonstrate more than 3 Gb/s using OFDM (Tsonev et al. 2014), and 15.73 Gb/s has been reported using off-the-shelf LEDs (Bian et al. 2019). These results use visible light and illumination-class devices and are the most representative of what LED-based Li-Fi can reach in the laboratory.

Infrared laser links. The figure of 224 Gb/s that is often cited for Li-Fi comes from a 2015 Oxford demonstration of an indoor bidirectional link over about 3 m, comprising six wavelength channels of 37.4 Gb/s each with a 60° field of view; a narrower 36° field of view supported three channels and 112 Gb/s (Gomez et al. 2015). The system used infrared laser sources, wavelength division multiplexing and beam steering, required a direct line of sight, and, at the time of the report, assumed a fixed receiver position (Savage 2015). A follow-up reported an aggregate rate of the order of 400 Gb/s (Gomez et al. 2016). These are important demonstrations of the capacity of optical wireless links, but they are not LED-based Li-Fi in the sense of a lighting-integrated, mobile, multi-user network.

The "100 times faster than Wi-Fi" claim. This ratio appears in vendor material, where the 224 Gb/s laboratory figure is compared with the roughly 7 Gb/s then attainable by the fastest Wi-Fi in the 60 GHz band (CNX Software 2023; pureLiFi n.d.). The comparison pairs a laboratory infrared link with a nominal RF figure and should not be read as a statement about typical user experience.

Standards and products. IEEE 802.11bb has a nominal peak of 9.6 Gb/s (Claburn 2023; pureLiFi 2023). The first compliant module is reported by its vendor to deliver more than 1 Gb/s, with a range of 20–300 cm, a 24° transmitter field of view and an operating wavelength of 850 nm (CNX Software 2023). In practical terms, early commercial Li-Fi therefore offers gigabit-class performance within a small cell, which is comparable to good Wi-Fi rather than far beyond it.

The more defensible argument for Li-Fi is therefore not peak speed but *aggregate capacity per unit area* and *latency and interference characteristics* in dense deployments, where small optical cells reuse spectrum that RF systems cannot (Haas 2018; Haas et al. 2016). Establishing this advantage quantitatively in field deployments, rather than by simulation or vendor statement, remains an open research task.

7. Applications and Maturity

Table 4 summarizes the application domains most often proposed for Li-Fi, the rationale in each, the evidence found in the sources reviewed and an indicative maturity level. Two domains merit additional comment.

Table 4 Application domains, supporting evidence and indicative maturity

Domain	Rationale	Evidence reviewed	Maturity
Healthcare	Avoids RF emission near sensitive equipment; high-rate data transfer	Proposals and discussion (Johnson 2023)	Early / pilot
Aircraft cabin	Reuses cabin lighting; no RF emission	One-off 12-seat flight trial, Oct. 2019 (Air France 2019; LEDs Magazine 2019)	Demonstration
Underwater	RF is strongly attenuated in water; blue-green optical window	Survey of underwater optical links (Zeng et al. 2017)	Research and prototypes
Offices, schools, secure rooms	Spatial confinement; extra capacity alongside Wi-Fi	Early products reported by vendors (CNX Software 2023; pureLiFi 2023)	Early commercial
Indoor positioning	Each luminaire identifies a small cell	Centimeter-level accuracy claimed by vendors (pureLiFi 2023)	Vendor-reported
Industrial and Internet of Things (IoT)	RF-hostile environments; predictable latency	Largely conceptual in the reviewed literature (Chowdhury et al. 2020)	Concept

Aviation. The first in-flight Li-Fi demonstration took place on 30 October 2019, on Air France flight AF6114, an Airbus A321 flying from Paris-Orly to Toulouse, with Latécoère's Li-Fi infrastructure installed on 12 seats (Air France 2019). It served a gaming tournament organized with Ubisoft. Reports at the time described it as a one-off proof of concept that would not immediately continue in service (LEDs Magazine 2019), and the airline stated that it was testing how readily the technology could be integrated and certified for commercial flights (Air France 2019). The trial therefore demonstrated feasibility, not a deployed service.

Underwater. Radio waves are strongly attenuated in water, whereas blue-green light propagates much further, which makes optical links attractive for short-range communication with remotely operated vehicles (Zeng et al. 2017). Such links are typically point-to-point optical systems and not networked Li-Fi in the sense defined in Section 2, but they share the same physical-layer principles.

The claim that Li-Fi is attractive for "green" networking deserves caution. Reusing luminaires avoids installing separate radio access points in some cases, but the Li-Fi front end, the backhaul and the uplink all consume power. A system-level energy comparison with Wi-Fi under equal-service conditions was not found in the sources reviewed and would be a useful contribution.

8. Limitations and Open Challenges

Table 5 lists the principal limitations together with mitigation strategies and research directions. Several of the limitations frequently quoted in introductory material need to be stated more carefully.

Table 5 Principal limitations of Li-Fi and corresponding mitigation or research directions

Challenge	Cause	Mitigation / research direction
Blockage and shadowing	Light does not penetrate opaque objects; the LOS path dominates	Distributed access points, diffuse NLOS paths, hybrid Li-Fi/Wi-Fi operation (Chowdhury et al. 2020; Komine et al. 2005)
Uplink	A user device cannot easily emit bright visible light	Infrared uplink or RF fallback (Haas et al. 2016)
Mobility and handover	Small optical cells	Multi-cell planning; inter-domain mobility in G.9991 Amd. 2 (Chowdhury et al. 2020; ITU-T 2019)
Ambient light	Sunlight and other luminaires add noise	Optical filtering, NIR operation, adaptive receivers (Matheus et al. 2019)
LED bandwidth	Slow phosphor response	Micro-LEDs, laser diodes, equalization, OFDM bit loading (Ferreira et al. 2016; Tsonev et al. 2014)
Dimming and flicker	The luminaire also provides illumination	Dimming-aware PHY modes; NIR operation (CNX Software 2023; IEEE 2018)
Cost and ecosystem	New luminaires, backhaul and client modules are needed	Standards-based interoperability (IEEE 2023; IEEE ComSoc Technology Blog 2023); client support is still limited
Security	Light leaks through windows and reflects from surfaces	Encryption and authentication still required (ITU-T 2019)

Weather. Introductory accounts often state that Li-Fi fails in bad weather. This applies to outdoor and FSO-type links, where fog, rain and turbulence degrade the signal (Mohsin and Murdas 2023), and not to indoor Li-Fi, where the relevant impairments are blockage and ambient light. Reviews should keep the two cases apart.

"No light, no connectivity." This limitation holds for visible-light systems that rely on the luminaire being on, although dimming to low levels is possible (Claburn 2023). It does not hold for NIR systems such as those developed for IEEE 802.11bb, which operate independently of room lighting (CNX Software 2023; pureLiFi 2023).

Security. Confinement of light to a room provides a physical-layer advantage, but light can leak through windows and be reflected, so confinement should be treated as a complement to, and not a substitute for, encryption and authentication. The 2020 amendment to ITU-T G.9991 adding IEEE 802.1X-based authentication reflects this (ITU-T 2019).

Ecosystem. Standardization removes a barrier to multi-vendor interoperability (IEEE ComSoc Technology Blog 2023), but a technology also needs client devices. To the author's knowledge, native Li-Fi support in mainstream smartphones and laptops has not yet become widespread, and Li-Fi is therefore most likely to be introduced first through dedicated modules and access points in specific environments.

9. Related Technology: Free-Space Optics

FSO is closely related to Li-Fi as a form of OWC, but it is a different application: narrow, invisible light beams link two fixed terminals over long distances, whereas Li-Fi uses room lighting to serve nearby

mobile devices. A recent example is Taara, a project that originated at X, Alphabet's "moonshot factory," and became an independent company in March 2025 (Converge Digest 2025; X 2025). Its Lightbridge terminals transmit at up to 20 Gb/s over distances of up to 20 km, and in February 2025 it introduced a silicon photonic chip that steers the beam with an optical phased array instead of mechanical parts; in outdoor tests two chips carried 10 Gb/s over 1 km (Krishnaswamy 2025). The relevance to Li-Fi lies in the shared technology base of optical beam steering and silicon photonics, which is also needed to improve receiver alignment and mobility support in indoor optical wireless systems (Gomez et al. 2016).

10. Research Directions and Outlook

Based on the literature reviewed, five directions appear most likely to determine whether Li-Fi moves beyond niche deployments.

- **Hybrid heterogeneous networks.** Li-Fi is most plausibly deployed as an additional layer alongside Wi-Fi and cellular networks, with intelligent selection and seamless handover between RF and optical links (Chowdhury et al. 2020; Haas 2018). The 802.11bb design of a light band that appears as another Wi-Fi band is consistent with this model (pureLiFi 2023).
- **Components.** Higher-bandwidth micro-LEDs and laser-based sources, together with receivers that combine a wide field of view with high gain, address the data-rate and mobility limits discussed above (Ferreira et al. 2016; Gomez et al. 2016; Tsonev et al. 2014). Laser arrays are a prominent candidate: simulation studies of VCSEL-based indoor access points report aggregate rates beyond 2 Tb/s per access point, but these are computed results and have not been demonstrated in measurements (Soltani et al. 2022).
- **Uplink and multi-cell design.** Low-power, eye-safe uplinks and practical cell planning and handover remain central to a usable network (Haas et al. 2016; ITU-T 2019).
- **Field measurements and system-level evaluation.** Most quantitative claims about capacity density, latency and energy come from simulations or vendor statements. Independent measurements in real buildings would strengthen the case.
- **Integration with the Internet of Things and machine learning.** Li-Fi's short range, spatial confinement and RF-free operation are consistent with secure, low-power links among IoT devices in settings such as hospitals and factories. In the author's view, machine-learning methods are also candidates for tasks such as interference management, handover prediction and traffic-aware resource allocation in dense multi-cell deployments. These ideas are prospective and, to the author's knowledge, have not yet been validated at scale.

11. Conclusion

Li-Fi has progressed from a laboratory concept and a 2011 demonstration to a standardized technology with early commercial products. The IEEE 802.11bb amendment, together with IEEE 802.15.7 and ITU-T G.9991, provides a basis for interoperable equipment. The headline data rates, however, need careful reading: the laboratory records of hundreds of gigabits per second were achieved with infrared lasers and beam steering, not with illumination LEDs, whereas the first standardized products operate at gigabit-class rates within a single small cell.

The evidence reviewed supports a more modest and more useful role than replacement of Wi-Fi: a short-range, high-capacity, RF-free layer for environments where interference, security or spectrum congestion matter. Realizing that role depends on solving practical issues, in particular uplink design,

multi-cell mobility, client device support and cost, and on independent field evidence of the capacity and energy benefits that are currently supported mainly by simulation and vendor statements.

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